

Machine Learning Interatomic Potentials

PWF Summer school on Theoretical and Computational Biophysics

University of Colombo

Day 5/5

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Where we left with AIMD

However, we have a timescale issue: individual PT events are fast, but rare on the AIMD timescale (tens of ps) → poor statistics on PT rates, mechanisms, etc... Enhanced sampling helps, but is expensive



Machine learning interatomic potentials

Goal

- We want to find a way to **speed up** simulations but **preserving the accuracy** of first-principles methods
- Learning the potential energy surface (PES) with a cheap procedure to run much longer simulations

What is a machine learning interatomic potential?

A machine learning interatomic potential (MLIP) is a model that predicts energy and forces from atomic positions, trained on reference data from quantum-mechanical calculations.



- The MLIP is a **black-box surrogate for DFT**: we feed in the atomic configuration (positions + elements), the model returns energy and forces directly, without solving the electronic structure at each step
- For periodic systems, MLIPs can also predict stresses (not a focus here)
- Supervised learning problem: every training configuration comes with a DFT label and a regression problem: outputs are continuous

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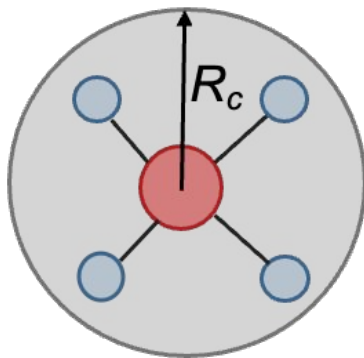


- **Total energy** is obtained as a sum of atomic energies (unlike DFT!)
- **Atomic forces** are obtained as derivatives: $F_i = -\nabla_i E$ (backpropagation) → this guarantees **conservative** forces and consistent energy–force predictions

Why does this work? The locality assumption

How can we decompose the total energy as a sum of local atomic contributions?

- For many systems, an atom's local environment is dominated by nearby neighbors.
- Each atom i is assigned an energy contribution ε_i based on its local chemical environment.
- The local environment contains only the neighboring atoms within a cutoff radius R_c .
- The total energy is obtained by summing all atomic contributions.

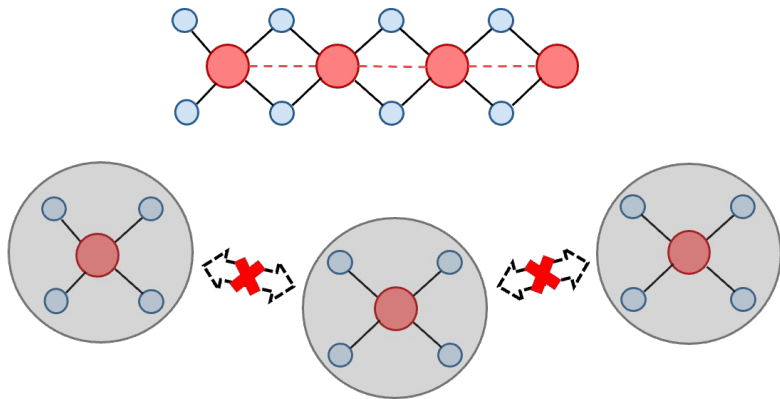


$$E_{tot} = \sum_i \varepsilon_i$$

Why does this work? The locality assumption

How can we decompose the total energy as a sum of local atomic contributions?

- **Transferability** → Same model, different system sizes: increasing the number of atoms, the sum over ϵ_i just has more terms.
- **Scalability** → Enables **linear scaling** with system size ($O(N)$ — unlike DFT).



Limitations

- Locality is an approximation → long-range interactions can extend beyond R_c .
- Standard MLIPs handle these poorly; extensions exist but add complexity.

Representing the local environment

For each atom i , we need to encode its local neighborhood as an input the model can process.

The representation must respect the physics:

- **Translational invariance** → moving the whole system by a constant vector should not change the energy.
- **Rotational invariance** → rotating the whole system should not change the energy.
- **Permutational invariance** → swapping two atoms of the same element should not change the energy.

Are raw Cartesian coordinates a good descriptor?

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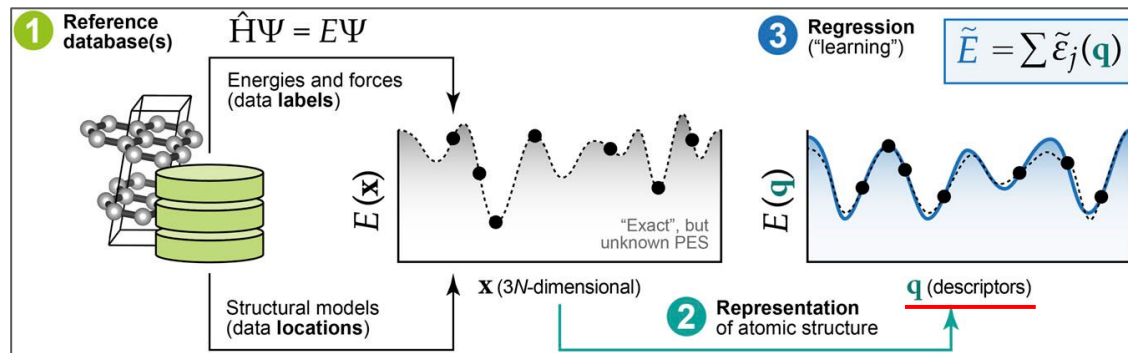


Are raw Cartesian coordinates a good descriptor?

Two families of approaches: **explicit** hand-designed descriptors, or **learned** representations.

Explicit descriptors: hand-designed features

- The atomic environment is transformed into a fixed-length vector of invariant features, chosen based on physical and chemical intuition.
- These functions form the input vector to a regression model that predicts the atomic energies



Examples of explicit **descriptors**:

Atom-Centered Symmetry Functions

Smooth Overlap of Atomic Positions

Hyperspherical Bispectrum Functions

Moment Tensor Functions

We define the descriptors → the model learns the energy from a predefined representation

Learned representations: features from data

Can the representation itself be learned from the atomic structure?

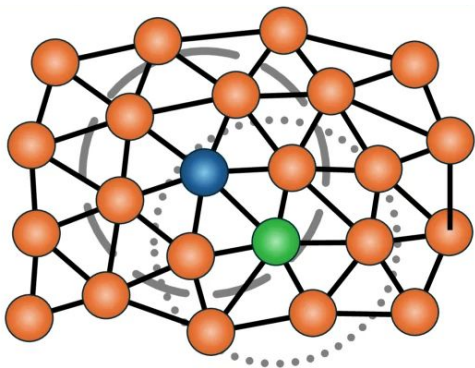
Atomic structure



Graph/local
environments



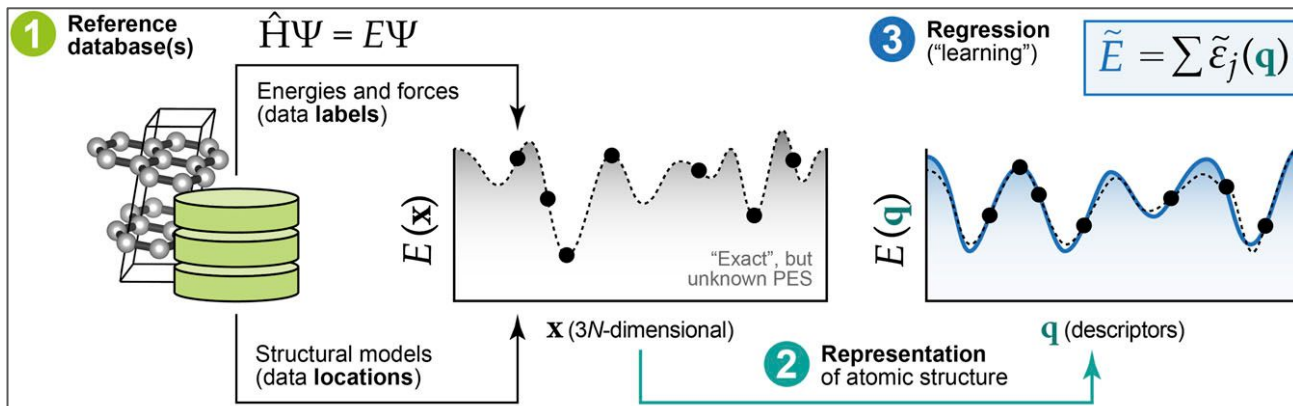
Learned atomic
features



- Atoms are represented as **nodes**; neighboring atoms within R_c are connected by **edges**.
- Information from neighboring atoms is combined iteratively $\rightarrow$ the atomic representation $\mathbf{h}_i$ is **learned during training**
- The architecture is designed to respect the required **translation, rotation and permutation symmetries**

Graph Neural Networks provide a natural framework for learning these representations.

Putting it together: MLIP workflow



- 1) Training data:** Structures, energies, and forces from reference electronic-structure calculations (typically DFT)
- 2) Representation:** Encode the local atomic environment while respecting the physical symmetries
- 3) Regressor:** Map the representation to atomic energy contributions $\epsilon_i \Rightarrow E_{tot} = \sum_i \epsilon_i$

Examples: neural networks, Gaussian process regression, linear regression

Foundational/universal MLIPs

Specialized MLIPs: trained for specific systems (elements, phases, thermodynamic states, ...)

Universal MLIPs: trained on very broad chemical and configurational spaces, aiming to transfer across many systems and conditions.

Semi-universal MLIPs: trained on subsets, e.g., organic molecules, inorganic materials, ...

Why they matter:

- Transferability across chemistry and phases → one model, many systems
- Avoid the DFT-data bottleneck → train once, reuse
- Zero-shot or few-shot deployment → new system can be tackled with little or no extra training data

= very cheap and user-friendly

Examples:

- M3GNet (2022) - trained on the Materials Project data (mainly inorganic materials)
- MACE-MP0 (2024) - MACE foundation model for materials chemistry
- MatterSim (2024) - Microsoft; materials across wide T and P ranges
- UMA (2025) - Meta FAIR; molecules + materials + catalysts

Foundational/universal MLIPs

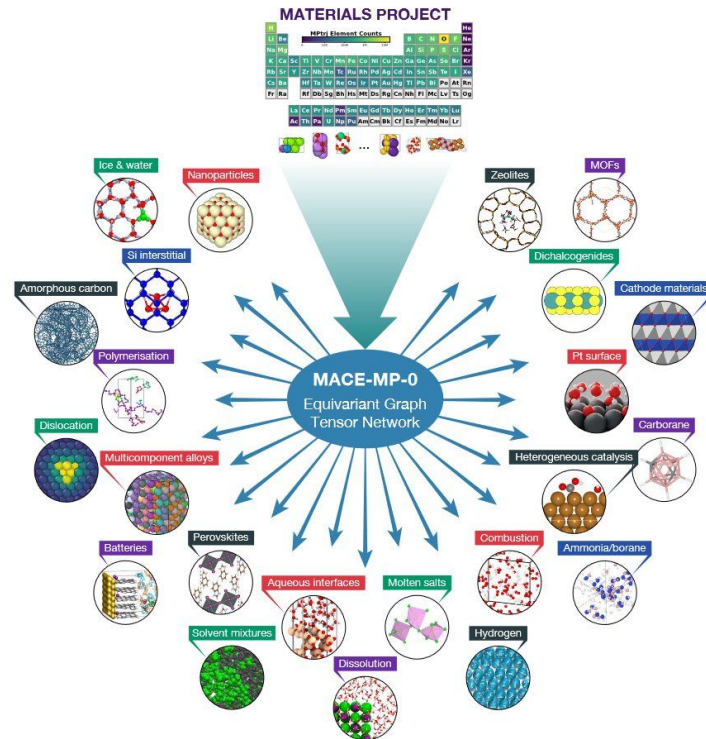
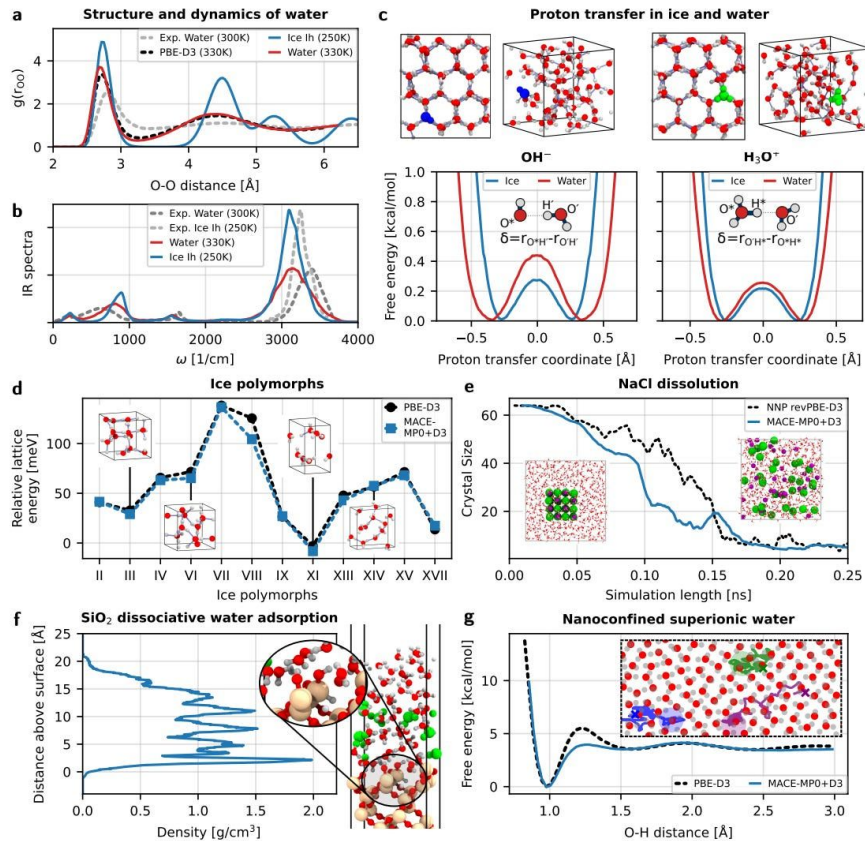


Figure 1: A foundation model for materials modelling. Trained only on Materials Project data (19) which consists primarily of inorganic crystals and is skewed heavily towards oxides, MACE-MP-0 is capable of molecular dynamics simulation across a wide variety of chemistries in the solid, liquid and gaseous phases.

Closing the loop: MLIPs for biophysics

This week we learnt:

- Classical MD → fast and useful for several applications, but no bond breaking nor electronic response
- Ab-initio MD (with DFT) → accurate physics, but limited to tens of ps
- MLIPs → near first-principles accuracy at much lower cost

What this unlocks:

- Rare events with better statistics (e.g., proton transfer)
- Possibility of studying larger systems for longer times
- Combined with enhanced sampling → free energies, rates, mechanisms

Closing the loop: MLIPs for biophysics

But keep in mind:

- MLIPs are only as good as their training data → reliability depends on how well the training set covers the configurations the simulation will visit
- Standard MLIPs still struggle with long-range interactions and excited-state processes — active research
- Universal MLIPs are impressive but not yet plug-and-play for every problem → **validation is still essential**

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Thank you for the attention!